

MA1521 L1 Week 01 Wed 20260812

div. by x^2 thru.

$$\text{Evaluate } \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 3}{3x^2 - 2x + 4} = \lim_{x \rightarrow \infty} \frac{1 + 2/x + 3/x^2}{3 - 2/x + 4/x^2} = \frac{\lim_{x \rightarrow \infty} (1 + 2/x + 3/x^2)}{\lim_{x \rightarrow \infty} (3 - 2/x + 4/x^2)}$$

$$= \frac{1 + 0 + 0}{3 - 0 + 0} = \boxed{\frac{1}{3}}$$

quot. rule

$$\text{Evaluate } \frac{d}{dx} \left(\frac{x^4 + 1}{x} \right) = \frac{(4x^3) \cdot x - (x^4 + 1) \cdot 1}{x^2} = \frac{3x^4 - 1}{x^2}$$

chain rule

der. of $2 \cos x$

$$\text{Evaluate } \frac{d}{dx} \ln(2 \cos x) = \frac{1}{2 \cos x} \cdot (-2 \sin x) = -\tan x$$

derivative of $\ln(-)$

$x+5-5 = 1 - \frac{5}{x+5} \Rightarrow \frac{dy}{dx} = \overset{\text{should be } +}{-5} \cdot (x+5)^{-2}$
 If $y = \frac{x}{x+5}$, what is $\frac{d^2y}{dx^2}$? $\Rightarrow \frac{d^2y}{dx^2} = \overset{\text{should be } -}{10} (x+5)^{-3}$

$(x > 0) \Rightarrow y = e^{(x \cdot \ln x)} \Rightarrow \frac{dy}{dx} = \underbrace{e^{(x \cdot \ln x)}}_{\text{chain rule}} \cdot \underbrace{(1 \cdot \ln x + x \cdot \frac{1}{x})}_{\text{prod rule}} = x^x \cdot (1 + \ln x)$
 If $y = x^x$, what is $\frac{dy}{dx}$?

If $xy - \ln y = 1$ and $xy \neq 1$, what is $\frac{dy}{dx}$?
 $\frac{d}{dx}(xy) \overset{\text{prod rule}}{=} 1 \cdot y + x \cdot \frac{dy}{dx}$
 $\frac{d}{dx}(\ln y) \overset{\text{chain rule}}{=} \frac{1}{y} \cdot \frac{dy}{dx}$

Implicit diff: diff both sides of eqn $xy - \ln y = 1$ to get $y + x \frac{dy}{dx} - \frac{1}{y} \frac{dy}{dx} = 0$
wrt x

so $(\frac{1}{y} - x) \cdot \frac{dy}{dx} = y$

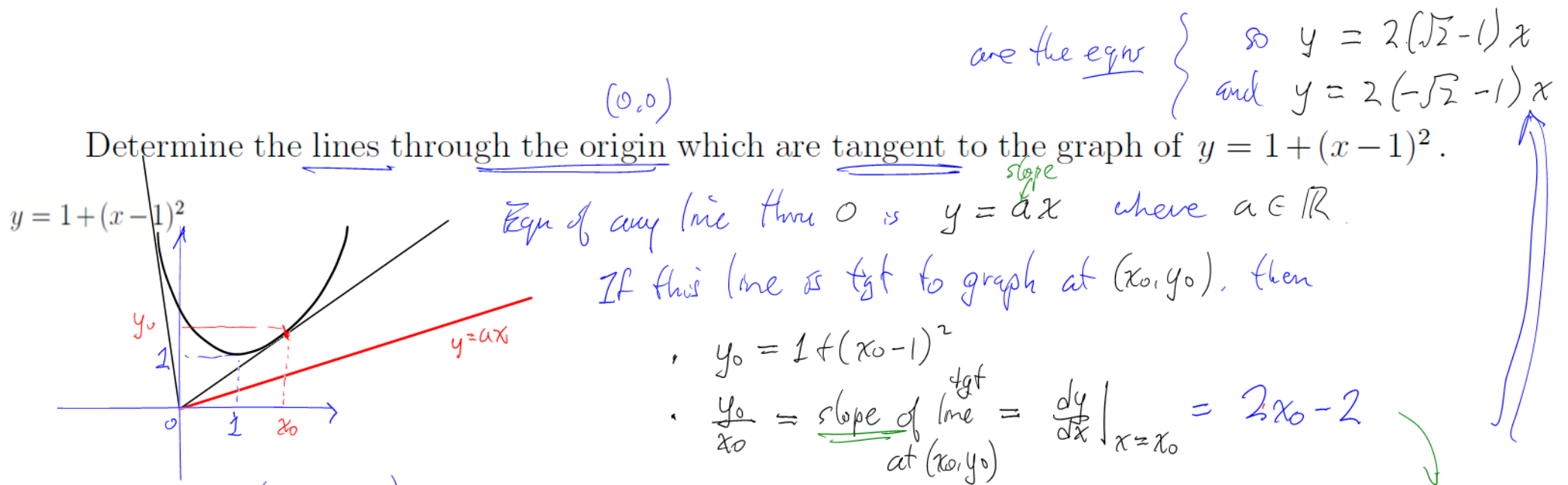
so $\frac{dy}{dx} = \frac{y^2}{1 - xy}$

If $xy - \ln y = 1$ and $xy \neq 1$, what is $\frac{d^2y}{dx^2}$?

Apply quot. rule: $\frac{d^2y}{dx^2} = \frac{\frac{d}{dx} \left(\frac{y^2}{1 - xy} \right)}{(1 - xy)^2} = \frac{\underbrace{2y \cdot \frac{dy}{dx}}_{\text{der of } y^2} \cdot (1 - xy) - y^2 \cdot \underbrace{(-1 \cdot y - x \cdot \frac{dy}{dx})}_{\text{der of } 1 - xy}}{(1 - xy)^2}$

$$= \frac{2y \cdot \frac{y^2}{1-xy} \cdot (1-xy) - y^2 \cdot \left(-y - x \cdot \frac{y^2}{1-xy}\right)}{(1-xy)^2}$$

$$= \frac{2y^3(1-xy) + y^3(1-xy) + xy^4}{(1-xy)^3} = \frac{y^3(3-2xy)}{(1-xy)^3}$$



Hence $(2x_0 - 2) \cdot x_0 = y_0 = 1 + (x_0 - 1)^2 \Rightarrow x_0^2 = 2$ so $x_0 = \pm\sqrt{2}$, with slope $\pm 2\sqrt{2} - 2$

What is the equation for the tangent line to the graph of $y = x^3 - x^2 + 5x - 6$ at $x = 1$?

At $x=1$, have $y=-1$, and $\frac{dy}{dx} \Big|_{x=1} = (3x^2 - 2x + 5) \Big|_{x=1} = 6$

$\Rightarrow$ eqn of tgt line is $(y - y_0) = (\text{slope}) \cdot (x - x_0)$

i.e. $y + 1 = 6(x - 1)$ i.e. $y = 6x - 7$

Evaluate $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x - \cos \frac{\pi}{2}}{x - \frac{\pi}{2}} \right) =$ defn of (derivative of $\cos x$)
^{ie $-\sin x$}
at $x = \frac{\pi}{2}$

Alt: use l'Hôpital's rule $= -1$

Evaluate $\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4} \stackrel{\text{l'Hôpital's rule}}{=} \lim_{x \rightarrow 0} \frac{-\sin x + x}{4x^3}$

$\stackrel{\text{again}}{=} \lim_{x \rightarrow 0} \frac{(-\cos x) + 1}{12x^2}$

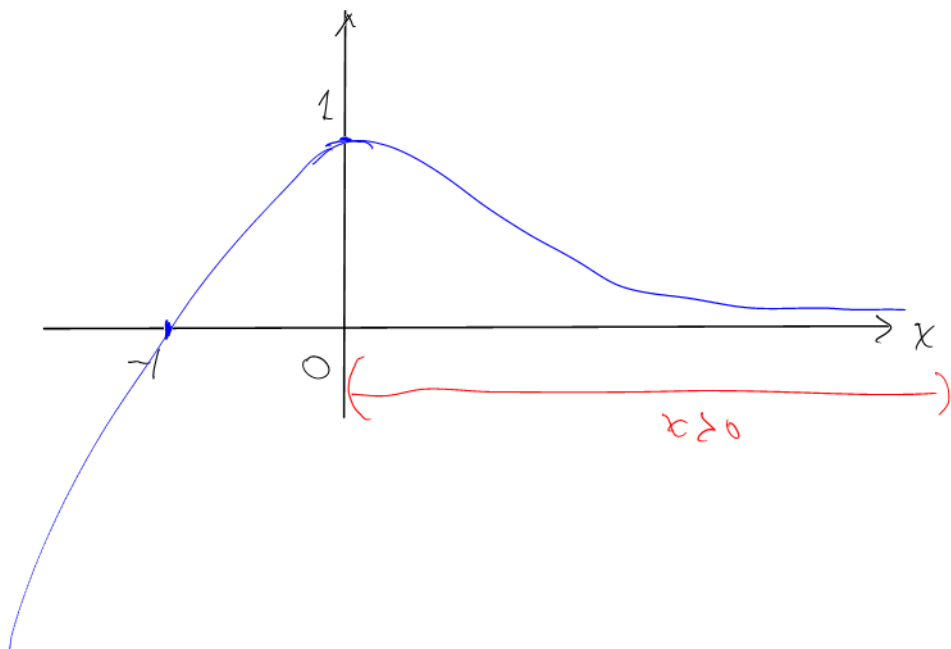
$\stackrel{\text{again}}{=} \lim_{x \rightarrow 0} \frac{\sin x}{24x}$

$\stackrel{\text{again}}{=} \lim_{x \rightarrow 0} \frac{\cos x}{24} = \boxed{\frac{1}{24}}$

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What is the minimum value of $(x+1)e^{-x}$ over all $x > 0$?

What is the maximum value of $(x+1)e^{-x}$ over all $x > 0$?



$$\text{Let } f(x) = (x+1)e^{-x}$$

$$\text{Then } f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$$

For $x > 0$, get $f'(x)$ is always < 0 strictly

$\Rightarrow f(x)$ is strictly decreasing

$\therefore$ no min value attained

As $x \rightarrow 0^+$ get $f(x) \rightarrow 1 =$

$\Rightarrow$ no max value attained

Suppose the function f satisfies

$$f(x+y) = f(x) + f(y) + x^2y + xy^2 \quad \text{for all real } x, y, \quad \text{and also } \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1.$$

Determine $f'(x)$.

Give the answer as an **expression** in terms of x .

Fix an arb. $x \in \mathbb{R}$. (Want to determine $f'(x)$)

$$\text{Recall defn. } f'(x) = \lim_{y \rightarrow 0} \left(\frac{f(x+y) - f(x)}{y} \right)$$

$$\begin{aligned} \text{Using relation } f(x+y) = f(x) + f(y) + x^2y + xy^2 &= \lim_{y \rightarrow 0} \left(\frac{f(y) + x^2y + xy^2}{y} \right) = \left(\lim_{y \rightarrow 0} \frac{f(y)}{y} \right) + \left(\lim_{y \rightarrow 0} x^2 \right) + \left(\lim_{y \rightarrow 0} xy \right) \end{aligned}$$

$$\begin{aligned} \lim_{y \rightarrow 0} \frac{f(y)}{y} &= 1 \\ &= 1 + x^2 + 0 \end{aligned}$$

$$\approx 1 + x^2$$

